

# Audion Image Tools

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[Русский](#) · [User Guide](#)

## Contents

- [Why It Exists](#)
- [Principles](#)
- [What It Can Do](#)
- [Next](#)
- [Technical Reference](#)
  - [Running](#)
  - [Defaults](#)
  - [Choosing a Colour Profile](#)
  - [Folders](#)
  - [Workbench Naming](#)

A portable toolkit for images: conversion, colour, sizing, cropping, watermarks, tiling, contact sheets.

## Why It Exists

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Working with images splits into two different jobs, and confusing them is expensive.

The first is **accept whatever you were given**. A shoot arrives as camera RAW, a scan as a multi-frame TIFF, a layout as PSD, a phone photo as HEIC. The program has to open all of it without asking where it came from.

The second is **deliver predictably**. Here the opposite matters: a narrow set of formats where behaviour is worked out in detail — what happens to transparency, how the colour profile is written, whether pixels change at all.

Hence the central decision: **wide on the way in, narrow on the way out**. On input, everything that can be opened. The deep, predictable processing is concentrated around JPG and PNG.

## Principles

**Three levels of format support are named aloud.** Not "supported", but how well.

| level       | what it means            | formats                                   |
|-------------|--------------------------|-------------------------------------------|
| stable core | worked out, always works | BMP, GIF, JPG, PNG, TGA, TIFF, WebP, AVIF |
| via adapter | needs an extra layer     | HEIC and HEIF, camera RAW — CR2, CR3, DNG |
| best effort | does not always open     | WMF, IFF, XIF; PSD as far as it goes      |

What is available on a particular machine, the program says itself:

```
python system_core\main.py formats
```

**Colour is chosen, not guessed.** sRGB for screens, CMYK for print — and in each case explicitly: the built-in path with no external file, or a specific profile from disk. The program will not decide which of the two CMYK profiles you need, because that depends on the print shop, not on the file.

**Operations that leave pixels alone are kept apart from those that don't.** Writing resolution changes only the tag in the file; fitting to a roll recalculates resolution from a size in millimetres while leaving the pixels as they are. This is stated at each such operation, so you don't later hunt for where the quality loss came from.

**Compatibility beats compactness.** WebP, AVIF, and HEIF are there for light delivery, but JPG remains the one that opens everywhere, and the program will not replace it with them by default.

## What It Can Do

| area       | about                                                                                 |
|------------|---------------------------------------------------------------------------------------|
| Conversion | to JPG, PNG, TIFF, WebP, AVIF, HEIF, HEIC; quality by preset or exact value           |
| Colour     | normalisation with an explicit profile, greyscale, safe transparency handling for JPG |
| Sizing     | to 1080p, 1440p, 2160p screens; to 16:9, A4, A3; to a roll by a side in millimetres   |
| Resolution | writing a chosen resolution without touching pixels                                   |
| Cropping   | a solid border, smart white-background crop, a safety margin in millimetres           |

| area           | about                                                                                       |
|----------------|---------------------------------------------------------------------------------------------|
| PDF assembly   | straight after cropping, with embedded PNG or JPG                                           |
| Watermark      | a caption in the corner or a diagonal protective line, with adjustable opacity and colour   |
| Tiling         | one image across A5, A4, A3, or a roll — for stickers, with margins and gaps in millimetres |
| Contact sheet  | a grid of thumbnails: standard sizes or your own                                            |
| TIFF splitting | multi-frame into numbered PNG frames                                                        |
| Rotation       | fixing by the orientation tag                                                               |

## Next

- [User Guide](#) — step by step.
- `tools\RELEASE_GUIDE_EN.md` — building a release.

## Technical Reference

### Running

```
launcher_gui.cmd      the main one, windowed
launcher_project.cmd  command line and text menu
launcher_project_ru.cmd the same in Russian
```

The menu uses the quick picker when available and falls back to a plain menu otherwise. For short, consequential choices it uses a simple list, so values cannot be confused at deeper steps.

### Defaults

| what                  | value                                                     |
|-----------------------|-----------------------------------------------------------|
| JPG quality           | 83; presets 60, 75, 90; exact values 1 to 100             |
| PNG compression       | one preset in the window, a parameter on the command line |
| contact sheet quality | 75 or 92                                                  |

| what             | value                      |
|------------------|----------------------------|
| diagonal caption | about 60 % of the diagonal |

## Choosing a Colour Profile

| task                                   | what to pick              |
|----------------------------------------|---------------------------|
| screen, website, ordinary delivery     | sRGB                      |
| print                                  | CMYK                      |
| built-in path without an external file | Pillow sRGB               |
| an explicit external reference         | the color.org profile     |
| broad classic print scenario           | Photoshop5DefaultCMYK.icc |
| coated print to a specific standard    | CoatedFOGRA39.icc         |

## Folders

```

input\      quick source folder
output\     results
logs\       run logs
report\     detailed reports
workspace\  temporary area
config\     defaults, themes, profile paths
system_core\ the implementation
install\    portable environment
wheelhouse\ packages for offline install
release\    release archives

```

## Workbench Naming

This project's vocabulary is the shared one across all Audion programs: **Source**, **Add file...**, **Target**, **Reset**, **Delete**, **List**. In Russian: **Источник**, **Добавить файл...**, **Назначение**, **Сбросить**, **Удалить**, **Список**.

**Reset** restores the project **input** and **output** without touching files. **Delete** clears the current source and target only after confirmation. The words **Destination**, **Clear**, «Цель», and «Очистить» are not used.